

A Review of Calculus: Sequences, Limits, Derivatives, and Integrals

N-20220319

§1.1 Sequences and convergence

After defining real numbers, the first analytic object to understand is a sequence. Intuitively,

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

should converge to 0, while

$$1, 2, 3, 4, \dots$$

should diverge. The difficulty is to turn this intuition into a definition that is precise.

Definition 1.1.1. A sequence is a function

$$a : \mathbb{N} \rightarrow \mathbb{R}$$

or more generally $a : \mathbb{N} \rightarrow \mathbb{C}$. We write a_n instead of $a(n)$, and denote the sequence by (a_n) or $(a_n)_{n=1}^{\infty}$.

Definition 1.1.2. Let (a_n) be a real sequence and let $\ell \in \mathbb{R}$. We say $a_n \rightarrow \ell$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that whenever $n \geq N$,

$$|a_n - \ell| < \varepsilon.$$

Equivalently,

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geq N) |a_n - \ell| < \varepsilon.$$

Example 1.1.3

Let

$$x_n = \frac{n}{n+1}.$$

Then $x_n \rightarrow 1$. Indeed,

$$|x_n - 1| = \frac{1}{n+1}.$$

Given $\varepsilon > 0$, choose $N > 1/\varepsilon - 1$. Then $n > N$ implies $|x_n - 1| < \varepsilon$.

§1.2 Limits of functions

Definition 1.2.1. Let $A \subseteq \mathbb{R}$, $f : A \rightarrow \mathbb{R}$, and a be a limit point of A . We write

$$\lim_{x \rightarrow a} f(x) = \ell$$

if

$$(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in A)(0 < |x - a| < \delta \Rightarrow |f(x) - \ell| < \varepsilon).$$

The condition $0 < |x - a|$ is important: the value of $f(a)$ is irrelevant, and f does not even have to be defined at a .

§1.3 Derivatives

Definition 1.3.1. The derivative of f at x is

$$\frac{df}{dx}(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists.

Theorem 1.3.2 (Chain rule)

If $f(x) = F(g(x))$, then

$$\frac{df}{dx} = \frac{dF}{dg} \frac{dg}{dx}.$$

Theorem 1.3.3 (Product rule)

If $f(x) = u(x)v(x)$, then

$$f'(x) = u'(x)v(x) + u(x)v'(x).$$

Theorem 1.3.4 (Quotient rule)

If $f(x) = u(x)/v(x)$, then

$$f'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{v(x)^2}.$$

Theorem 1.3.5 (Leibniz rule)

If $f = uv$, then

$$f^{(n)}(x) = \sum_{r=0}^n \binom{n}{r} u^{(r)}(x) v^{(n-r)}(x).$$

§1.4 Small-o and big-O

Definition 1.4.1. As $x \rightarrow x_0$, write $f(x) = o(g(x))$ if

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

Write $f(x) = O(g(x))$ if $f(x)/g(x)$ remains bounded near x_0 .

Thus $o(g)$ means much smaller than g , while $O(g)$ means no larger than a constant multiple of g . Clearly,

$$f = o(g) \Rightarrow f = O(g).$$

§1.5 Integration

An integral is a limit of sums. The usual Riemann integral is modeled by

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{n=0}^N f(x_n) \Delta x.$$

For an interval partition with $\Delta x = (b-a)/N$ and $x_n = a + n\Delta x$, this approximates the area under the graph.

If f is differentiable, then over a small interval the area is

$$f(x_n) \Delta x + O((\Delta x)^2).$$

Summing and letting $N \rightarrow \infty$ gives the usual integral.

Theorem 1.5.1 (Fundamental theorem of calculus)

Let

$$F(x) = \int_a^x f(t) dt.$$

Then, under the usual continuity hypothesis on f ,

$$F'(x) = f(x).$$

Remark 1.5.2 — This is a review note rather than a full proof-based analysis text. The next revision should add precise hypotheses for every theorem, especially differentiability and continuity assumptions.

§1.6 Why the epsilon definition is shaped this way

The definition of convergence has the order of quantifiers

$$\forall \varepsilon > 0 \exists N \forall n \geq N.$$

This order is essential. We are not allowed to choose one fixed tolerance; the sequence must eventually enter every tolerance window around the limit. The number N may depend on ε , because smaller error demands later terms.

For example, to prove $1/n \rightarrow 0$, given $\varepsilon > 0$, choose $N > 1/\varepsilon$. Then $n \geq N$ implies

$$\left| \frac{1}{n} - 0 \right| < \varepsilon.$$

The proof is not a calculation of the limit; it is a demonstration that every requested precision can eventually be met.

§1.7 Continuity by sequences

A function f is continuous at a if

$$x_n \rightarrow a \implies f(x_n) \rightarrow f(a)$$

for every sequence (x_n) in the domain. This sequence criterion is often easier to use than epsilon-delta directly. To prove discontinuity, it suffices to find one sequence $x_n \rightarrow a$ such that $f(x_n) \not\rightarrow f(a)$.

For example, the Dirichlet function is discontinuous everywhere. Near any a , choose rational and irrational sequences both converging to a . Their function values approach different limits.

§1.8 Derivative as best linear approximation

The derivative is often introduced as a slope:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

A more structural reading is that differentiability means

$$f(a+h) = f(a) + f'(a)h + o(h).$$

Thus the derivative is the coefficient of the best linear approximation near a . This viewpoint generalizes directly to multivariable calculus, where the derivative becomes a linear map.

§1.9 Integral as accumulation

The definite integral can be understood as the limit of accumulated small contributions. For continuous f , Riemann sums

$$\sum_i f(\xi_i) \Delta x_i$$

converge to the area-signed accumulation. The fundamental theorem of calculus says that differentiation and integration are inverse operations in the sense that

$$\frac{d}{dx} \int_a^x f(t) dt = f(x).$$

This is not merely an area formula; it says local rate and global accumulation determine one another.

§1.10 Convergence as topology

The elementary epsilon definition in this note is the first appearance of a topology. On the real line, saying

$$x_n \rightarrow x$$

means that every interval around x eventually contains all x_n . In a metric space (X, d) the same sentence becomes

$$\forall \varepsilon > 0, \quad \exists N, \quad n > N \Rightarrow d(x_n, x) < \varepsilon.$$

Thus the elementary sequence definition is not a special trick for numbers; it is the definition of convergence once a notion of distance has been chosen.

This immediately explains why different spaces have different convergence. In $C[0, 1]$ with the sup norm,

$$\|f_n - f\|_\infty = \max_{x \in [0, 1]} |f_n(x) - f(x)| \rightarrow 0$$

means uniform convergence. With the L^1 norm,

$$\|f_n - f\|_1 = \int_0^1 |f_n(x) - f(x)| dx \rightarrow 0,$$

thin spikes may disappear in integral size while still being tall. The same elementary sentence “goes to zero” therefore depends on the surrounding space.

§1.11 Derivative as a linear map

In one-variable calculus, the derivative is introduced as a number:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

The higher-level meaning is that $f'(a)h$ is the best first-order linear approximation to the change in f :

$$f(a+h) = f(a) + f'(a)h + o(h).$$

For a map $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the derivative at a is a linear map $DF(a)$ satisfying

$$F(a+h) = F(a) + DF(a)h + o(|h|).$$

The Jacobian matrix is only the coordinate representation of this linear map.

§1.12 Limits and derivatives test topology and linearization

Elementary limit and derivative computations operate on two layers. At the calculation layer, they train epsilon estimates and algebraic simplification. At the structural layer, they introduce topology and linearization. A limit asks whether a function is continuous with respect to the chosen topology; a derivative asks whether the function is locally a linear map plus a smaller error.